

JAVA EXPERIMENT-1

SUBMITTED BY:-

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1. Explore and understand the role of JDK, JRE, and JVM.

➤ JDK-:

JDK stands for Java Development Kit, it is a software development kit used to write and compile Java programs. It includes tools like javac (compiler), java (runtime), and libraries for Java development.

➤ JRE-:

JRE stands for Java Runtime Environment, it provides the libraries, Java virtual machine and other components to run Java applications. It is a subset of JDK.

➤ JVM-:

JVM is Java Virtual Machine, it runs Java bytecode on your machine. It converts the bytecode (compiled java code) into machine code for execution. It ensures platform independence.

2. Install JDK and verify Java environment.

Visit the Oracle JDK website.

Go to the macOS section and download the desired version of JDK.

JDK Development Kit 21.0.1 downloads

JDK 21 binaries are free to use in production and free to redistribute, at no cost, under the Oracle No-Fee Terms and Conditions (NFTC).

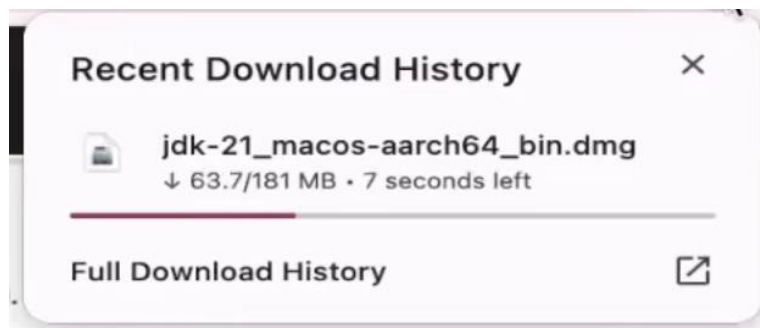
JDK 21 will receive updates under the NFTC, until September 2026, a year after the release of the next LTS. Subsequent JDK 21 updates will be licensed under the Java SE OTN License (OTN) and production use beyond the limited free grants of the OTN license will require a fee.

Linux **macOS** Windows

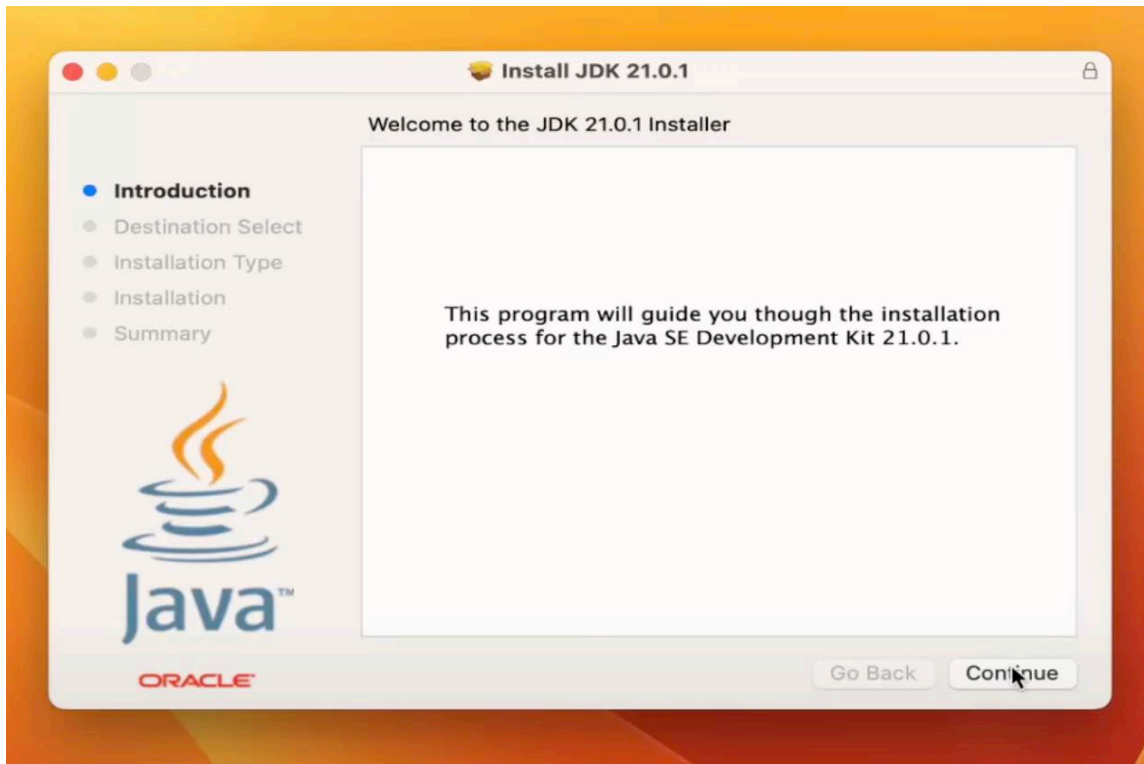
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ARM64 Compressed Archive	181.92 MB	https://download.oracle.com/java/21/latest/jdk-21_macos-aarch64_bin.tar.gz (sha256)
ARM64 DMG Installer	181.26 MB	https://download.oracle.com/java/21/latest/jdk-21_macos-aarch64_bin.dmg (sha256)
x64 Compressed Archive	184.17 MB	https://download.oracle.com/java/21/latest/jdk-21_macos-x64_bin.tar.gz (sha256)
x64 DMG Installer	183.50 MB	https://download.oracle.com/java/21/latest/jdk-21_macos-x64_bin.dmg (sha256)

[Documentation Download](#)

It will start downloading.



We need to follow all the next steps to download the JDK.



After the installation, open terminal and type this command to see that our JDK is successfully downloaded.

```
dakshseghal@Dakshs-MacBook-Air ~ % /usr/libexec/java_home -V
Matching Java Virtual Machines (2):
 21.0.6 (arm64) "Oracle Corporation" - "Java SE 21.0.6" /Library/Java/JavaVirtualMachines/jdk-21.jdk/Contents/Home
 17.0.13 (x86_64) "Eclipse Adoptium" - "OpenJDK 17.0.13" /Library/Java/JavaVirtualMachines/temurin-17.jdk/Contents/Home
dakshseghal@Dakshs-MacBook-Air ~ %
```

3. Create a simple hello word program and execute it.

CODE-:

```
1  public class hello {  
    Run | Debug  
2      public static void main(String[] args) {  
3          System.out.println(x:"Hello, World!");  
4      }  
5  }  
6  
7
```

OUTPUT-:

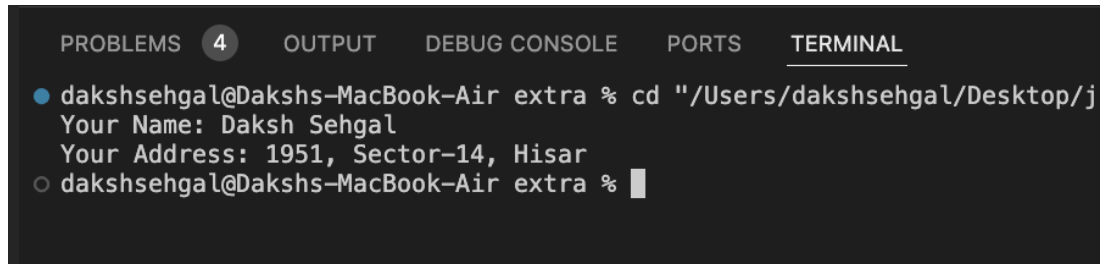
```
PROBLEMS 3 OUTPUT DEBUG CONSOLE PORT  
cd "/Users/dakshseghal/Desktop/java codes/ext  
● dakshseghal@Dakshs-MacBook-Air java codes % c  
Hello, World!  
○ dakshseghal@Dakshs-MacBook-Air extra %
```

4. Display your name and address in different line.

CODE-:

```
1  public class display {  
    Run | Debug  
2      public static void main(String[] args) {  
3  
4          System.out.println(x:"Your Name: Daksh Sehgal");  
5  
6          System.out.println(x:"Your Address: 1951, Sector-14, Hisar");  
7      }  
8  }  
9
```

OUTPUT-:



A screenshot of a terminal window with a dark background. At the top, there is a navigation bar with five tabs: 'PROBLEMS', '4', 'OUTPUT', 'DEBUG CONSOLE', 'PORTS', and 'TERMINAL'. The 'TERMINAL' tab is selected and underlined. Below the tabs, the terminal shows a command prompt 'dakshsehgal@Dakshs-MacBook-Air extra %' followed by the command 'cd "/Users/dakshsehgal/Desktop/j'. The output of the command is displayed on the next line: 'Your Name: Daksh Sehgal' and 'Your Address: 1951, Sector-14, Hisar'. The prompt then changes to 'dakshsehgal@Dakshs-MacBook-Air extra %' with a cursor at the end.

```
PROBLEMS 4 OUTPUT DEBUG CONSOLE PORTS TERMINAL
● dakshsehgal@Dakshs-MacBook-Air extra % cd "/Users/dakshsehgal/Desktop/j
  Your Name: Daksh Sehgal
  Your Address: 1951, Sector-14, Hisar
○ dakshsehgal@Dakshs-MacBook-Air extra % █
```